

AI Research Agent — Project Architecture

Project purpose

AI Research Agent is designed as an evidence-grounded single-agent research application. The user supplies a research topic; the system discovers sources, retrieves original webpages, ranks source quality, asks a CrewAI research agent to analyze the evidence, and uses Groq to synthesize a structured report.

Deployment model

The project is designed for direct GitHub upload and Streamlit Community Cloud deployment. No local environment, .env file, or local API-key configuration is required.

Final workflow

User topic → Research queries → DDGS discovery → Source ranking → Webpage retrieval → CrewAI research → Evidence package → Groq synthesis → Cited report → Source list

Repository

```
ai-research-agent/  
■■■ app.py  
■■■ config.py  
■■■ research_agent.py  
■■■ research_tools.py  
■■■ source_quality.py  
■■■ report_generator.py  
■■■ prompts.py  
■■■ utils.py  
■■■ requirements.txt  
■■■ README.md  
■■■ .gitignore  
■■■ .python-version  
■■■ .streamlit/  
■■■ config.toml
```

Security

The Groq API key must exist only in Streamlit Secrets. It must never be committed to GitHub, README files, screenshots or source code. The application reads GROQ_API_KEY and GROQ_MODEL from Streamlit Secrets.

Research quality model

Tier	Typical sources	Role
Tier 1	Government, official institutions, major international organizations, academic/research domains	Highest priority
Tier 2	Universities, research organizations, professional organizations	Strong supporting evidence
Tier 3	Established publications and industry sources	Context and secondary evidence
Tier 4	General websites/blogs	Discovery only unless independently verified

Quality principles

Search results are treated as discovery, not proof. The system attempts to retrieve the original webpage before using the information. Important claims should be traceable to a source. Conflicting findings should be disclosed. Lack of evidence should result in an explicit limitation rather than a fabricated answer.

Professional UI

The Streamlit interface includes a research topic area, research-depth control, report-type selection, target source count, evidence-recency preference, live workflow status, source metrics, expandable source evidence, and Markdown report download.